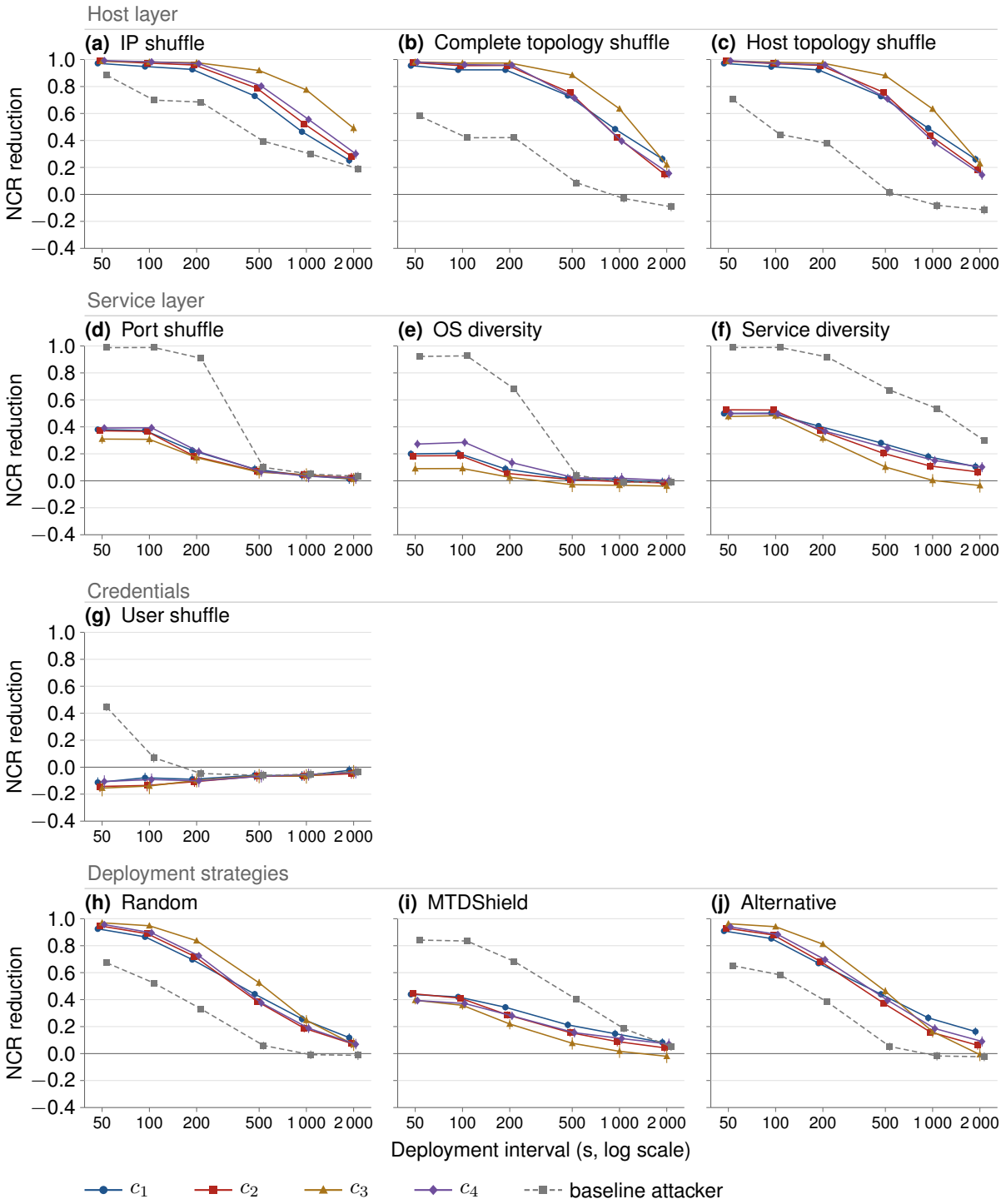

Host layer
(a) IP shuffle
(b) Complete topology shuffle
(c) Host topology shuffle
Service layer
(d) Port shuffle
(e) OS diversity
(f) Service diversity
Credentials
(g) User shuffle
Deployment strategies
(h) Random
(i) MTDShield
(j) Alternative
NCR reduction
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
50
100
200
500
1 000
2 000
Deployment interval (s, log scale)
c_1
c_2
c_3
c_4
baseline attacker